

Problem Set 1

Assignment Details

Due Date: September 3, 2026

Total points: 50

Name: _____

Problem 1: Algebra (15 pts)

Solve the following system of equations for (x, y) or (x, y, z) :

1.

$$y = x + 4$$

$$y = 3 - x$$

2.

$$x = 5y + 1$$

$$y = 7x - 4$$

3.

$$y = 3x + 4$$

$$y = 3x + 5$$

4.

$$4y = 8x - 4$$

$$x = y + 1$$

5.

$$y = 5x - 2z$$

$$y = 2x + 5$$

$$z = x$$

Problem 2: Derivatives (20 pts)

Take the derivative of each function with respect to its independent variable(s).

1.

$$y = 4x - 5$$

2.

$$y = 2x^2 + 3x - 4$$

3.

$$x = t^3 - 4t + 1$$

4.

$$\frac{dy}{dt} = t + t^2$$

5.

$$y = 2e^{-3t}$$

6.

$$y = (1 - x)^2$$

7.

$$y = \ln(x^2)$$

8. Take the derivative of x and y separately.

$$f(x, y) = x^a y^b$$

9.

$$f(x) = \frac{x - a}{x^2}$$

10.

$$f(x) = [ax^{\frac{1}{1-c}} + bx^{\frac{1}{1-c}}]^{1-c}$$

Problem 3: Optimization (15 pts)

Suppose you have a function

$$\pi(q) = pq - \frac{1}{a}q^2,$$

which represents your firm's profits as a function of quantity produced q . Profits consist of two components: revenue and costs. Identify which term represents revenue and which represents costs. Recalling what you know about derivatives, find the value of q that maximizes profits q^* and its associated maximum profits π^* .